Why is the residual $\boldsymbol{z}$ uncorrelated with $\boldsymbol{x}_1$?

Supplementary note for AS4501 (Gaussian process derivation)

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Setting

Let $\boldsymbol{x} = (\boldsymbol{x}_1, \boldsymbol{x}_2)^T$ be jointly Gaussian,

$$\boldsymbol{x} \sim \mathcal{N}\left(\begin{pmatrix}\boldsymbol{\mu}_1\\ \boldsymbol{\mu}_2\end{pmatrix}, \Sigma = \begin{pmatrix}\Sigma_{11} & \Sigma_{12}\\ \Sigma_{21} & \Sigma_{22}\end{pmatrix}\right), \qquad \Sigma_{12} = \Sigma_{21}^T.$$

In the Gaussian-process posterior derivation we used the following identity, here stated for completeness:

$$\boldsymbol{x}_2 \mid \boldsymbol{x}_1 \;\sim\; \mathcal{N}\Big(\boldsymbol{\mu}_2 + \Sigma_{21}\Sigma_{11}^{-1}(\boldsymbol{x}_1 - \boldsymbol{\mu}_1),\;\; \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}\Big).$$

The proof of this identity hinges on the claim that the *residual*

$$\boldsymbol{z} \;=\; \boldsymbol{x}_2 \;-\; \Sigma_{21}\Sigma_{11}^{-1}\,\boldsymbol{x}_1 \tag{1}$$

is uncorrelated with $\boldsymbol{x}_1$. The purpose of this note is to spell out *why* that is true.

Algebraic verification

Using bilinearity of covariance,

$$\begin{aligned}\mathrm{Cov}(\boldsymbol{z}, \boldsymbol{x}_1) &= \mathrm{Cov}\big(\boldsymbol{x}_2 - \Sigma_{21}\Sigma_{11}^{-1}\boldsymbol{x}_1,\; \boldsymbol{x}_1\big) &(2)\\ &= \mathrm{Cov}(\boldsymbol{x}_2, \boldsymbol{x}_1) \;-\; \Sigma_{21}\Sigma_{11}^{-1}\,\mathrm{Cov}(\boldsymbol{x}_1, \boldsymbol{x}_1). &(3)\end{aligned}$$

Substituting the known block-covariance entries $\mathrm{Cov}(\boldsymbol{x}_2, \boldsymbol{x}_1) = \Sigma_{21}$ and $\mathrm{Cov}(\boldsymbol{x}_1, \boldsymbol{x}_1) = \Sigma_{11}$:

$$\mathrm{Cov}(\boldsymbol{z}, \boldsymbol{x}_1) = \Sigma_{21} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{11} = \Sigma_{21} - \Sigma_{21} = \mathbf{0}. \tag{4}$$

The algebra is short, but the choice of coefficient $\Sigma_{21}\Sigma_{11}^{-1}$ looked unmotivated. The next two sections explain why this is precisely the right coefficient.

Where the coefficient comes from

Look for the matrix A that makes $\mathrm{Cov}(\boldsymbol{x}_2 - A\boldsymbol{x}_1,\; \boldsymbol{x}_1) = 0$. Expanding,

$$\mathrm{Cov}(\boldsymbol{x}_2 - A\boldsymbol{x}_1,\; \boldsymbol{x}_1) = \Sigma_{21} - A\,\Sigma_{11} = \mathbf{0} \quad\Longleftrightarrow\quad A = \Sigma_{21}\,\Sigma_{11}^{-1}.$$

There is a *unique* linear combination of $\boldsymbol{x}_1$ that, when subtracted from $\boldsymbol{x}_2$, removes all linear dependence. That is the content of equation (**??**).

Geometric intuition

Consider the inner product on random variables given by $\langle u, v\rangle := \operatorname{Cov}(u, v)$. With this inner product, *uncorrelated* means *orthogonal.*

The vector $A\boldsymbol{x}_1 = \Sigma_{21}\Sigma_{11}^{-1}\boldsymbol{x}_1$ is the *orthogonal projection* of $\boldsymbol{x}_2$ onto the linear span of the components of $\boldsymbol{x}_1$. The residual $\boldsymbol{z}$ is, by definition, the component of $\boldsymbol{x}_2$ perpendicular to that span. The Pythagorean decomposition reads

$$\boldsymbol{x}_2 \;=\; \underbrace{\Sigma_{21}\Sigma_{11}^{-1}\,\boldsymbol{x}_1}_{\text{projection}} \;+\; \underbrace{\boldsymbol{z}}_{\perp\,\boldsymbol{x}_1}.$$

The same fact appears in ordinary least squares: residuals are orthogonal to the regressors.

Connection to linear regression

Given centred $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$, the least-squares regression coefficient

$$A^\star \;=\; \arg\min_A \mathbb{E}\big[\|\boldsymbol{x}_2 - A\boldsymbol{x}_1\|^2\big]$$

has the closed form $A^\star = \Sigma_{21}\Sigma_{11}^{-1}$. So the construction of $\boldsymbol{z}$ is exactly: *subtract the best linear predictor of $\boldsymbol{x}_2$ given $\boldsymbol{x}_1$.* Whatever is left is, by the defining optimality, uncorrelated with the regressor. This is the “magic” that makes the Gaussian conditioning identity work.

Why uncorrelated $\Rightarrow$ independent (Gaussian only)

For *arbitrary* distributions, uncorrelated does not imply independent. For *jointly Gaussian* variables it does. Since $\boldsymbol{z}$ is an affine function of $\boldsymbol{x} = (\boldsymbol{x}_1, \boldsymbol{x}_2)$, the pair $(\boldsymbol{z}, \boldsymbol{x}_1)$ is again jointly Gaussian. A jointly Gaussian pair with zero covariance has joint density that factors into a product of marginals, hence the variables are independent. This is what lets us write $\boldsymbol{z} \mid \boldsymbol{x}_1 = \boldsymbol{z}$ in the GP proof and read off the conditional moments by substituting $\boldsymbol{x}_2 = \boldsymbol{z} + \Sigma_{21}\Sigma_{11}^{-1}\boldsymbol{x}_1$:

$$\mathbb{E}[\boldsymbol{x}_2 \mid \boldsymbol{x}_1] = \boldsymbol{\mu}_2 + \Sigma_{21}\Sigma_{11}^{-1}\,(\boldsymbol{x}_1 - \boldsymbol{\mu}_1), \tag{5}$$

$$\operatorname{Var}(\boldsymbol{x}_2 \mid \boldsymbol{x}_1) = \operatorname{Var}(\boldsymbol{z}) \;=\; \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}. \tag{6}$$